AP PHYSICS 1

Unit 1 • Worksheet 1.5A • Answer key

Worksheet 1.5A • Vectors and Motion in Two Dimensions

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A ball is launched at 25 m/s at 37° above the horizontal ($\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$). What is the horizontal component of its velocity?

(A) 15 m/s

(B) 20 m/s

(C) 25 m/s

(D) 12.5 m/s

Solution

Answer: (B). $v_x = 25\cos 37^\circ = 25(0.8) = 20\,\text{m/s}$.

Question 2

A vector has components $A_x = 6$ and $A_y = -8$. What is its magnitude?

(A) 14

(B) 2

(C) 10

(D) 48

Solution

Answer: (C). $\sqrt{6^2 + 8^2} = 10$.

Question 3

A ball is thrown horizontally at 8 m/s from the top of a cliff and lands 2 s later. How high is the cliff?

(A) 20 m

(B) 16 m

(C) 10 m

(D) 40 m

Solution

Answer: (A). Vertical: $h = \frac{1}{2}gt^2 = \frac{1}{2}(10)(4) = 20\,\text{m}$. The horizontal speed does not matter.

Question 4

How far from the base of the cliff does the ball in the previous question land?

(A) 8 m
(B) 4 m
(C) 32 m
(D) 16 m

Solution
Answer: (D). Horizontal: $x = v_x t = 8(2) = 16\,\text{m}$.

Question 5

At the same moment and from the same height, ball 1 is dropped and ball 2 is thrown horizontally. Which lands first?

(A) Ball 1
(B) Ball 2
(C) They land at the same time.
(D) It depends on their masses.

Solution
Answer: (C). Both start with $v_y = 0$ and have the same vertical acceleration g, so they fall the same height in the same time.

Question 6

What is the acceleration of a projectile at the highest point of its path?

(A) Zero
(B) g downward
(C) g upward
(D) Horizontal

Solution
Answer: (B). Gravity acts during the whole flight: $a = g$ downward, even at the top.

Question 7

A ball is launched at 30 m/s at 53° above level ground ($\cos 53° = 0.6$, $\sin 53° = 0.8$). How long is it in the air?

(A) 2.4 s
(B) 4.8 s
(C) 3 s
(D) 6 s

Solution
Answer: (B). $v_{0y} = 24\,\text{m/s}$; time to the top $24/10 = 2.4\,\text{s}$; on level ground the total time is twice that: 4.8 s.

Question 8

Projectiles are launched with the same speed from level ground. Which launch angle gives the greatest range?

(A) 30°
(B) 60°
(C) 45°
(D) 90°

Solution
Answer: (C). Without air resistance the range is greatest at 45° (30° and 60° give the same, smaller range).

Question 9 (Free response)

A soccer ball is kicked from the ground at $20\,\text{m/s}$ at 37° above the horizontal ($\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$).

(a) Find the horizontal and vertical components of the initial velocity.
(b) Find the time to reach the highest point and the maximum height.
(c) Find the total time in the air and the horizontal range.
(d) What is the speed of the ball at its highest point?

Solution
(a) $v_x = 16\,\text{m/s}$, $v_{0y} = 12\,\text{m/s}$. (b) $t = 12/10 = 1.2\,\text{s}$; $h = \frac{12^2}{2(10)} = 7.2\,\text{m}$. (c) $2.4\,\text{s}$; $16 \times 2.4 = 38.4\,\text{m}$. (d) Only v_x remains: $16\,\text{m/s}$.